DO NOW — RECALL

Unfamiliar programming challenges

Name: _____ Class: _____ Date: _____

1. What is a loop invariant?

2. What distinguishes a run from a distinct value?

3. Why handle empty arrays explicitly?

4. For a=[1,1,2,1], give count of distinct values and runs.

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

A property true before and after each completed iteration.

2. Model answer

Runs count contiguous groups; distinct counts globally unique values.

3. Model answer

Otherwise a[0] could be invalid and initial count wrong.

4. Model answer

Distinct values 2; runs 3 ([1,1],[2],[1]).

By the end, you can…

• A run is a maximal group of equal consecutive values, not a unique value across the whole array.

• Maintain invariant: after index i, the count equals the number of runs in the processed prefix.

• Handle empty input and test all-equal, alternating and separated-repeat arrays.

Vocabulary — know these terms

run — Consecutive equal-value group.

invariant — Property maintained across iterations.

edge case — Boundary or unusual input.

distinct — Different values irrespective of position.

NEW LEARNING

Read, annotate and explain

Understand

A run is a maximal group of equal consecutive values, not a unique value across the whole array.

Apply

Maintain invariant: after index i, the count equals the number of runs in the processed prefix.

Evaluate

Handle empty input and test all-equal, alternating and separated-repeat arrays.

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

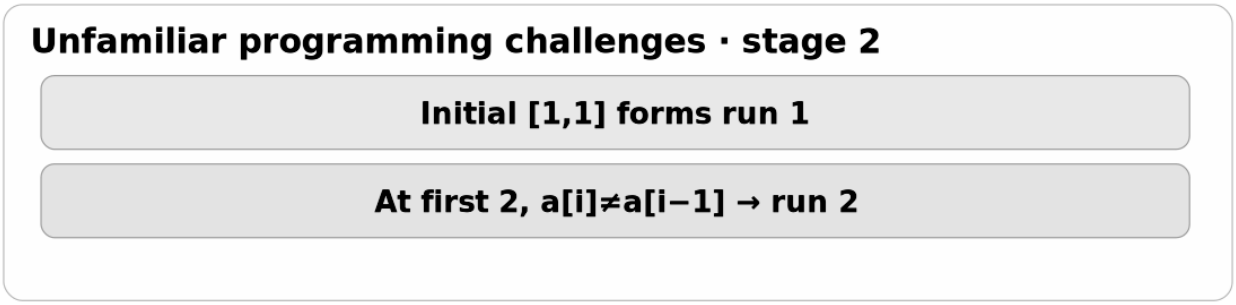
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Unfamiliar programming challenges · stage 1: Initial [1,1] forms run 1

New Learning



Unfamiliar programming challenges · stage 2: Initial [1,1] forms run 1; At first 2, a[i]≠a[i−1] → run 2

New Learning



Unfamiliar programming challenges · stage 3: Initial [1,1] forms run 1; At first 2, a[i]≠a[i−1] → run 2; Second 2 still run 2

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



Unfamiliar programming challenges · stage 4: Initial [1,1] forms run 1; At first 2, a[i]≠a[i−1] → run 2; Second 2 still run 2; Final 1 differs → run 3

I DO — WORKED EXAMPLE

Follow the worked method

Trace CountRuns on [4,4,5,5,5].

Predict first

Consecutive repeated fives belong to the same second run.

Explain and check

Explain the reason for the result.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace CountRuns on [4,4,5,5,5].

Consecutive repeated fives belong to the same second run.



IDO · stage 1: Consecutive repeated fives belong to the same second run.

Trace CountRuns on [4,4,5,5,5].

Consecutive repeated fives belong to the same second run.



IDO · stage 2: Consecutive repeated fives belong to the same second run.; Explain the reason for the result.

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

We Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace CountRuns on [1,2,1,2].

There are four runs but only two distinct values.



WEDO · stage 1: There are four runs but only two distinct values.

WE DO — TRY · DISCUSS · IMPROVE

TRY: Use the original guided questions and diagrams in the preceding We Do pages. DISCUSS: Explain and challenge methods before independent practice. IMPROVE: Note one change now; compare with the source model only AFTER the Exit Ticket.

Guided 1: Predict first There are four runs but only two distinct values.

DISCUSS — Compare your methods. Justify one important decision, identify one difference, and agree a specific improvement.

IMPROVE — Write one precise correction or stronger explanation in your own answer.

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 1

Trace CountRuns for [3,3,4,4,3,5,5]. State where runs increment. [6]

Question 2

Write a valid C# CountRuns(int[] a) returning runs and handling empty input. [8]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations, a trace or justified reasoning as appropriate.

Question 3

For [1,2,1], compare runs and distinct counts, and assess time/extra-space complexity of CountRuns. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking the answer.

Runs always equal number of unique values.

How confident are you? Circle one: Confident / Getting there / Need help

Explain your choice, identifying one concept you can now use independently.

GREEN PEN — Correct and explain

Choose one mistake from your independent work. Write the correct method and explain precisely why your original answer was wrong.

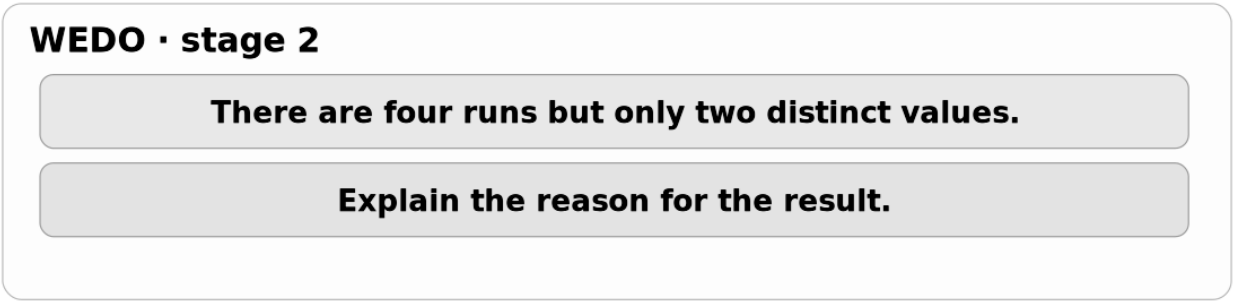
Next step: revisit the model or a relevant revision-guide section if needed.

We Do Check · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace CountRuns on [1,2,1,2].

There are four runs but only two distinct values.



WEDO · stage 2: There are four runs but only two distinct values.; Explain the reason for the result.

WE DO — SOURCE CHECK / IMPROVE

Only use this section AFTER attempting the We Do, You Do and Exit Ticket. Compare with the original classroom lesson's CHECK explanation. Correct errors in GREEN PEN and explain why.

Guided 1 — original CHECK

Explain and check

Explain the reason for the result.

IMPROVE: Explain one correction to your own reasoning.